



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: August 3rd, 2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No	: CE F211
Course Title	: Mechanics of Solids
Instructor-in-Charge	: G MUTHUKUMAR (muthug@pilani.bits-pilani.ac.in , 1209-H)
Team of instructors	: Anshuman (Prof.), (anshu@pilani.bits-pilani.ac.in , 1211 – E) Mukund Lahoti (Prof.) (mukund.lahoti@pilani.bits-pilani.ac.in , 1210 - D) KUMAR RISHABH (RS) <p20240022@pilani.bits-pilani.ac.in>, Research Scholar's room DEINA SUSAN SONY (RS) <p20250402@pilani.bits-pilani.ac.in>, Research Scholar's room

1. Course Description: Introduction to mechanics of rigid bodies and deformable bodies, Thermal stresses, Equilibrium of forces, Bending moment and shear force diagrams for determinate beams and frames, Analysis of statically determinate trusses; Flexural and shear Stresses in beams, combined stresses, Stresses and strains on inclined planes, Introduction to torsion, Torsion in shafts, Slope and deflection in beams due to bending, Introduction to Energy Methods, Stresses in thin cylindrical shells, Suspension cables, Failure theories, Buckling of columns using Euler's Theory.

2. Scope and Objective of the Course: This course aims at providing overall understanding of various methods in understanding the behavior of structural elements under various loading conditions. As a part of this course, understanding of behavior of key structural materials will also be touched upon. Also, an overview of key structural components and their importance to structure will also be covered.

3. Course learning outcomes:

- To comprehend the basic understanding of behavior of structural elements
- To get a detailed understanding of different methods of approaching the problem
- To gain the complete knowledge of stresses and strains developed in structural elements
- To develop the comprehensive capability to analyze the structural elements

4. Text Books & Reference Books:

TB1: S. H. Crandall et al., An Introduction to Mechanics of Solids, McGraw-Hill International editions.

RB1: Mechanics of Materials; F. P. Beer, E. R. Johnston, J. T. DeWolf and D. F. Mazurek, Fifth Edition, 2011, McGraw-Hill Education Pvt. Ltd., India.

RB2: Engineering Mechanics of Solids, Egor P. Popov, 2nd Edition, 1998, PHI, India.

RB3: Mechanics of Materials, James M. Gere, 6th Edition, 2006, Thomson Books/Cole, USA

RB4: Engineering Mechanics (Statics and Dynamics), N N Dubey, McGrawHill

5. Method of conducting the course:

- Both blackboard and PPT will be used during the lecture session
- Lecture notes will be uploaded on Nalanda for reference.

6. Course Plan:

Week	Module No.	Lecture No.	Lecture Session	Reference	Learning outcomes
1	1	1-3	Introduction to Mechanics of Solids • Fundamentals of Principles of Mechanics • Concept of Force • Moment of Force • Equilibrium conditions • Introduction to Friction • Free Body Diagram	Chapter 1	Overview
2	2	4-6	Introduction to Mechanics of deformable bodies • Analysis of deformable bodies • Uniaxial loading and deformation • Statically determinate situation • Statically indeterminate situation • Analysis of trusses • Castiglianos' theorem (Application)	Chapter 2	Overview mechanics bodies
3-4	3	7-14	Forces and Moments transmitted by slender members • Forces and Moments at a section • Distributed loads • Resultants of distributed loads	Chapter 3	Overview forces and moments members

			- Differential equilibrium relationships - Singularity functions		
5-6	4	15-20	Stress and strain - Stress & Plane stress - Differential element in plane stress - Stress components (arbitrarily oriented faces in plane stress) - Mohr's circle representation of plane stress/general state of stress - Analysis of deformation - Definition of strain components - Plain strain & relationship between strain and displacement - Strain components (arbitrary sets of axes) - Mohr's circle representation of plane strain/general state of strain - Measurement of strains	Chapter 4	Comprehensive understanding of stress & strain
7-8	5	21-25	Stress-strain temperature relations - Tensile test - Idealization of stress-strain curve - Elastic stress-strain relations - Thermal strain - Complete equations of elasticity - Complete Elastic Solution for a Thick-walled Cylinder - Strain energy in an elastic body - Stress concentration - Composite materials and anisotropic elasticity - Criteria for initial yielding - Introduction to plastic stress-strain	Chapter 5	Comprehension of stress-strain relationship
9-10	6	26-30	Torsion Geometry of deformation of a	Chapter 6	Overview of torsion

			twisted circular shaft • Stresses obtained from stress-strain relations • Stress and deformation in a twisted elastic circular shaft • Torsion of elastic hollow circular shafts • Stress analysis in torsion, combined stresses • Strain energy due to torsion • Yielding in torsion		
11-12	7	31-35	Stresses due to bending • Geometry of deformation of a symmetrical beam subjected to pure bending • Stresses obtained from stress-strain relations • Equilibrium requirements • Stress and deformation in symmetrical elastic beams (pure bending) • Stresses in symmetrical elastic beams transmitting both shear force and bending moment • Stress analysis in bending (combined stresses) • Strain energy due to bending • Yielding in bending	Chapter 7	Overview of stresses due to bending
13	8	36-39	Deflections due to bending • Moment-curvature relation • Integration of the moment-curvature relation • Superposition • Load-deflection differential equation • Energy methods	Chapter 8	Comprehensive understanding of deflections due to bending
14	9	40-42	Stability of equilibrium buckling • Elastic stability • Elastic stability of flexible columns • Elastic post-buckling behavior • Instability as a mode of failure	Chapter 9	Introduction to buckling

7. Evaluation Scheme:

Component	Weightage	Duration	Date & Time	Comments
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	(%)			
Tutorial 1	5%	50 minutes (each)	August 17 th , 2026 (Tut hour)	Open book
Tutorial 2	5%	50 minutes (each)	September 7 th , 2026 (Tut hour)	Open book
Tutorial 3	5%	50 minutes (each)	September 21 st , 2026 (Tut hour)	Open book
Tutorial 4	5%	50 minutes (each)	October 26 th , 2026 (Tut hour)	Open book
Tutorial 5	5%	50 minutes (each)	November 16 th , 2026 (Tut hour)	Open book
Mid-Semester Examination	35%	1.5 hours	<TEST_1>	Closed book
Comprehensive Examination	45%	3 hours	<TEST_C>	Closed book

Criterion for NC:

- If the candidate misses one particular component completely, NC will be awarded.
- 30% of the average of the top three performers or 40% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.

Note: Best 4 out of 5 Tutorials will be considered (20% weightage)

8. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components. However, there is no minimum percentage linking attendance and assessment.

9. Chamber Consultation Hour: Wednesday (6 PM – 7 PM).

10. Make-up Policy: Makeup will be given only for genuine cases (accompanied by proof) for Mid-Sem or comprehensive exam only.

11. Note (if any): Notice, if any, concerning the course will be displayed on the Nalanda LMS.

**Instructor-in-charge
CE F211**